Step 7 : Decide on H_0 choosing the suitable rejection rule from the following table corresponding to H_1.

Alternative Hypothesis (H_1)	$\mu_X \neq \mu_Y$	$\mu_X > \mu_Y$	$\mu_X < \mu_Y$
Rejection Rule	$\lvert t_0 \rvert \geq t_{n-1,\frac{\alpha}{2}}$	$t_0 > t_{n-1,\alpha}$	$t_0 < -t_{n-1,\alpha}$

Example 2.3

The following table gives the scores (out of 15) of two batches of students in an examination.

Batch I	6	7	9	2	13	3	4	8	7	11
Batch II	5	6	5	7	1	7	2	7		

Test at 1% level of significance the average performance of the students in Batch I and Batch II are equal.

Solution:

Step 1 : **Hypotheses:** Let μ_X and μ_Y denote respectively the average performance of students in Batch I and Batch II. Then the null and alternative hypotheses are :

Null Hypothesis $H_0 : \mu_X = \mu_Y$

i.e., the average performance of the students in Batch I and Batch II are equal.

Alternative Hypothesis $H_1 : \mu_X \neq \mu_Y$

i.e., the average performance of the students in Batch I and Batch II are not equal.

Step 2 : **Data**

The given sample information are:

Sample size for Batch I : m =10

Sample size for Batch II : n = 8

Step 3 : **Level of significance**

α = 1%

Step 4 : **Test statistic**

The test statistic under H_0 is

$$T = \frac{\overline{X} - \overline{Y}}{S_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$$

The sampling distribution of T under H_0 is the t-distribution with $m+n-2$ degrees of freedom *i.e.*, $t \sim t_{m+n-2}$

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